Omar Sharaf

Director, Pricing Model Engineering & MLOps

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SUMMARY

Engineering leader with 8+ years at Fortune 100 insurance and healthcare companies, specializing in taking actuarial and data science models into governed, production-grade deployment. Currently Director of Data Engineering for Pricing Model Support at Travelers, leading teams that doubled the number of models in production and delivered four Generative AI applications to production in 18 months.

CORE SKILLS

Leadership: Engineering Management, Hiring & Team Building, Mentorship & Coaching, Agile / SAFe, Product Ownership, Stakeholder Partnership

AI/ML: Generative AI, RAG, LLM Applications, MLOps, Real-time & Batch Model Deployment

Cloud & Data: AWS, Databricks, Snowflake, Terraform, Airflow, Docker, Neo4j, MongoDB, Teradata, Hadoop

Languages & Tools: Python, SQL, TypeScript, JavaScript, R, CI/CD, Tableau, Angular

EXPERIENCE

Director of Data Engineering — Pricing Model Support

Feb 2026 – Present

The Travelers Companies

Hartford, CT (Hybrid)

- Partner with key stakeholders, data scientists, and actuaries to build the roadmap for model deployments, serving as their trusted engineering partner and coaching engineers across lines of business to raise engineering standards.
- Established governed, production-grade delivery within two weeks on an inherited team with no testing or release controls: CI/CD pipelines, code-quality gates, and production-release controls.
- Hired four engineers directly and lead a team of five, mixing junior and senior engineers.
- Rolled out Claude Code workflows across the full delivery lifecycle (deployment, validation, and test generation) to speed up development for junior and senior engineers alike, with a human in the loop at every stage: engineers review and approve AI output, and data science signs off on validation.
- Cut a critical production job from up to 8 hours to 30 minutes, turning chronically missed SLAs into consistently met ones.
- Reduced an 11-hour model run to 1 hour 40 minutes, letting the business run models and make decisions during the workday instead of overnight.
- Converted a data scientist's prototype into production-grade code in under 24 hours using Claude Code, validated and approved by data science.
- Introduced Agile sprints and act as product owner, defining epics and features with the business while staying hands-on in code.

Technical Engineering Manager — Gen AI / MLOps

Aug 2024 – Feb 2026

The Travelers Companies

Hartford, CT (Hybrid)

- Led four teams spanning data products, AI intake pipelines, and Gen AI applications: 11 direct reports, 3 borrowed engineers embedded on the team, and about 8 offshore developers reporting into those teams. Owned delivery, performance reviews, and technical direction after a leadership transition.
- Doubled the number of ML models in production, clearing the full pricing backlog until data science had no models left to deploy.
- Built reusable deployment patterns that cut model-to-production time to under 24 hours with no added headcount.
- Delivered four Gen AI applications to production in 18 months across RAG and AI intake use cases, including the company's first Gen AI application released to external brokers.
- Served on the hiring panel that selected 10–15 high-performing engineers.
- Drove adoption of GitHub Copilot and early AI tooling, improving team efficiency and engineers' technical depth.

Cloud Data Engineer — Gen AI / MLOps Mar 2024 – Aug 2024
The Travelers Companies Hartford, CT (Hybrid)

- Led development and deployment of real-time and batch machine-learning models on the company's MLOps platform, ensuring scalability and reliability.
- Orchestrated production deployments of models using Terraform and Airflow, keeping releases stable across environments.
- Designed and implemented monitoring for deployed ML models to detect and resolve issues proactively.
- Served as lead developer on the company's first RAG application, built on GPT-3.5.
- Partnered with business stakeholders to set technical priorities and translate business requirements into technical roadmaps.
- Mentored junior engineers on MLOps practices, Python development, and cloud architecture, and streamlined onboarding for permanent and contract hires.

Data Engineer Oct 2022 – Mar 2024
The Travelers Companies Hartford, CT (Hybrid)

- Built a Python tool that converts SQL Server table DDLs into the SQL of any target platform (Snowflake, Databricks, and others) and detects schema changes in SQL Server, applying them to the target automatically, so data could live in both places and stay in sync without manual migration work.
- Developed a Neo4j-backed API and web frontend that visualizes database hierarchy and improves data visibility for engineering teams.
- Created an automated testing framework and set Snowflake deployment standards across the line of business.
- Stepped in as interim lead for two teams during a leadership transition.

Data Engineer Jun 2021 – Sep 2022
The Cigna Group (CGI Inc consultant) Bloomfield, CT

- Designed Python automation tools and frameworks that shortened development cycles and saved hundreds of engineering hours annually.
- Built ETL procedures that extract and transform base-table data for downstream applications, enforcing database development standards.
- Developer on a system-migration project, integrating a new upstream data feed into the warehouse: used Airflow to automate moving on-prem data, joining it with cloud data, and loading it into Teradata Vantage, so downstream customers kept receiving data in the exact same formats with no visible change.
- Delivered incrementally in SAFe Agile teams.

Programmer Analyst Jun 2018 – Jun 2021
The Cigna Group (CGI Inc consultant) Bloomfield, CT

- Built Python REST APIs serving data from Hadoop, Teradata, and Oracle, and delivered business-critical reporting through Hadoop ELT and Tableau.
- Developed Angular applications in Docker containers and maintained their CI/CD pipelines.

EDUCATION

M.S., Data Science (in progress), Central Connecticut State University Jan 2022 – Present
B.A., Economics, University of Connecticut Aug 2014 – Dec 2019

CERTIFICATIONS

AWS Certified Cloud Practitioner, Amazon Web Services Apr 2021